Homework 1 · Part A

Formative mock examination · Lectures 1 and 2

Time: 120 minutes. Diagnostic total: 100 points.

Attempt this part in one timed sitting before using the training part or solutions. Write by hand on paper. For an honest diagnostic, use only this question paper, without notes, internet or AI. This is a practice recommendation, not an announcement of the actual final exam's permitted materials.

The TA will mark this part seriously, but mistakes here do **not** lower your final course grade. Keep your original attempt; label any corrections made after the timed sitting. Its diagnostic score is separate from the assessed Part B score.

True/false: 10 claims, 4 points each	40
Two tickets, one weekend	20
One more lesson, one less coffee	30
Two short answers, 5 points each	10
Total	100

True/false instructions. State whether each claim is true or false and justify your answer. For a true claim, provide an argument; for a false claim, give a fully specified counterexample. An unsupported label earns no credit.

Rational preferences are complete and transitive. Continuity means that weak comparisons survive limits. Convex preferences have convex upper contour sets. Strong monotonicity means $x \geq y$, $x \neq y \Rightarrow x \succ y$; $x \gg y$ means every coordinate is strictly larger. Neighbourhoods are relative to the stated domain.

A1. True or false · Lecture 1

1. There is always a winner

On $X = \mathbb{R}^2_+$, define

$$x \succeq y \quad \Longleftrightarrow \quad x = y \text{ or } x_1 > y_1 \text{ or } x_2 > y_2.$$

These preferences are rational.

2. A jump in the score

On $X = \mathbb{R}^2_+$, preferences are represented by

$$u(x) = x_1 + x_2 + \mathbf{1}\{x_1 + x_2 \geq 1\},$$

where $\mathbf{1}$ is the indicator function. These preferences are continuous.

3. Mixing without a concave index

On $X = \mathbb{R}^2_{++}$, preferences are represented by

$$u(x) = x_1^2 x_2^2.$$

These preferences are strictly convex.

4. Somewhere better means everywhere higher

On $X = \mathbb{R}^2_+$, rational, continuous, convex and locally nonsatiated preferences must satisfy

$$x \gg y \quad \Rightarrow \quad x \succ y.$$

5. A hilltop or just a landing?

Let $X = \mathbb{R}^2_+$. A continuous utility function u represents convex preferences. If x^* is a local maximiser of u, then it is also a global maximiser.

Here, a local maximiser means that $u(x^*) \geq u(x)$ for every feasible x in some neighbourhood of x^*; the inequality need not be strict.

A1. True or false · Lecture 2

For questions 6–10, $X = \mathbb{R}^2_+$, $p \gg 0$, $m > 0$, and

$$B(p,m) = \{x \in X : p \cdot x \le m\}, \qquad D(p,m) = \arg\max_{x \in B(p,m)} u(x).$$

The Kuhn–Tucker convention is

$$\mathcal{L} = u(x) + \lambda(m - p \cdot x) + \mu_1 x_1 + \mu_2 x_2, \qquad \lambda, \mu_1, \mu_2 \ge 0.$$

6. A different currency, the same shopping basket

For every $t > 0$,

$$D(tp, tm) = D(p, m),$$

even when the consumer has several optimal bundles.

7. Strict curvature, a unique choice

If a continuous utility function u is strictly convex, then the consumer has a unique optimal bundle on every budget set.

8. All boxes ticked

Suppose a continuously differentiable utility function represents convex, strongly monotone preferences. Every feasible bundle satisfying the Kuhn–Tucker conditions is a global solution to the consumer's problem.

9. Left on the shelf

Suppose u is differentiable, strictly concave and strongly increasing. If the unique optimal bundle satisfies $x_1^* > 0$ and $x_2^* = 0$, then

$$\frac{u_2(x^*)}{p_2} < \frac{u_1(x^*)}{p_1},$$

where u_i denotes the partial derivative with respect to good i.

10. Can a smooth value hide several choices?

Suppose u is continuous and preferences are locally nonsatiated. Let

$$v(p,m) = \max_{x \in B(p,m)} u(x).$$

If v is differentiable at (p,m) and $v_m(p,m) > 0$, then the optimal consumption bundle at (p,m) is unique.

A2. Two tickets, one weekend · 20 points

Mira and Anton are planning a cultural trip. A programme contains x_1 hours of live music and x_2 hours of exhibitions. Their satisfaction scores are

$$s_M(x) = 2x_1 + x_2, \qquad s_A(x) = x_1 + 3x_2.$$

Their friend organising the trip wants neither traveller to be left behind: she ranks programmes by the *lower* of the two scores. Her preferences on $X = \mathbb{R}^2_+$ are therefore represented by

$$u(x) = \min\{s_M(x), s_A(x)\}.$$

(a) Rank the programmes $(3,0)$, $(0,3)$ and $(1,1)$. Are the organiser's preferences complete, transitive and continuous? Explain.

(b) Draw the indifference set at utility level 6 and its weak upper contour set. Label the intercepts and the point where the indifference set changes slope. Explain what happens to the identity of the less satisfied traveller at this point.

(c) Are the organiser's preferences convex? Strictly convex? Strongly monotone? Locally nonsatiated? Justify each answer.

(d) The booking platform replaces u with $v = u^2$ when displaying programme scores. Does this preserve the organiser's preferences? Is v concave as a function?

(e) Could *any convex utility function* represent the organiser's preferences? Prove your answer.

A3. One more lesson, one less coffee · 30 points

A student is preparing for an examination and can purchase q hours of individual tutoring at a price $p > 0$ per hour. Let a denote money left for all other purchases. The first lessons help a great deal, but each additional hour is less valuable. The student's preferences are represented by

$$u(a, q) = a + 6\log(1 + q), \qquad (a, q) \in \mathbb{R}^2_+.$$

The student has income $m > 0$ and cannot borrow. The price of the other-purchases good a is 1, so the budget is

$$a + pq \leq m.$$

(a) Does an optimal bundle exist? Must it exhaust the budget? Is it unique? Justify your answers.

(b) Write the Lagrangian and all Kuhn–Tucker conditions, including the nonnegativity constraints.

(c) Derive demand for both goods for every $p > 0$ and $m > 0$. Identify the parameter regions in which the student buys no tutoring, spends everything on tutoring, or consumes both goods. Include the boundaries between these regions.

(d) Let $p = 2$. Find the optimal bundle and the budget multiplier at incomes $m = 2$ and $m = 8$. Explain why additional income changes tutoring hours in one case but not the other.

(e) In the region where both goods are consumed, derive indirect utility $v(p, m)$. Recover tutoring demand using Roy's identity. Explain the economic meaning of v_m.

A4. Two short answers · 10 points

Answer *each* question in no more than three sentences. Each question is worth 5 points. Treat these as hypothetical cases; no knowledge of a particular news article is required.

1. Paying to resist temptation

A person pays for an app that blocks social media, even though they could simply choose not to open it for free.

Does paying to remove an available option necessarily reveal irrational preferences? Explain which assumption of the standard consumer-choice model matters for your answer.

2. Too cheap to be desirable

A shopper wants a bag costing 100,000 rubles but says they would lose interest if the same model cost 10,000 rubles and became widely available.

Does this contradict rational consumer choice? Explain what the shopper might be consuming besides the bag's physical characteristics and what would need to remain unchanged for a meaningful comparison.

Homework 1 · Part B

Training problem set · Lectures 1 and 2

This is the regular assessed homework part, organised as a step-by-step training set. Work at your own pace, use the lecture notes and practise the calculations. Write the submitted solution by hand on paper. There is no time limit for this part.

The assessed total is 100 points, independent of the diagnostic Part A score. The three assessed homeworks are equally weighted within the 20% homework component of the course grade. Cite external sources and disclose any AI assistance; you must be able to explain all submitted work. Submission dates and procedures are announced by the TA.

Show the intermediate steps requested below. In every optimisation problem, distinguish existence, uniqueness and the calculation of a candidate. Unless stated otherwise, quantities are divisible and nonnegative, prices are strictly positive, and income is positive.

B1. A preference checklist · 15 points

On $X = \mathbb{R}^2_+$, consider

$$u_1 = 2x_1 + x_2, \quad u_2 = \min\{x_1, 2x_2\}, \quad u_3 = -(x_1 - 1)^2 - (x_2 - 2)^2.$$

(a) Make a table stating whether each represented relation is complete, transitive and continuous. Explain the common argument behind your answers.

(b) Add columns for convexity of preferences, strict convexity of preferences, strong monotonicity and local nonsatiation. Give one short argument or counterexample for each classification.

(c) Draw the indifference sets through $(2, 2)$ for all three utilities. Label the utility levels, and shade the corresponding weak upper contour sets.

B2. Representations and group indices · 10 points

Let $u(x) = x_1 + 2x_2$ on $\mathbb{R}^2_+$.

(a) Which of $3u - 7$, u^2 and $\min\{u, 2\}$ represent the same preferences? Explain using the range of u; give two bundles witnessing any failed representation.

(b) Draw the indifference sets $u = 2$ and $u = 4$. Relabel them under the first two transformations without changing the curves.

(c) Now consider $F(t, c, z) = (1 + \sqrt{tc})(1 + z)$, with $t, c, z \geq 0$. Write $F = U(v(t, c), z)$ with U strictly increasing in v. Explain why the ranking of two (t, c) bundles at a common z does not change when that common z changes. Does this determine a fixed spending budget for the group?

B3. Budget geometry · 10 points

Prices are $p = (2, 3)$ and income is $m = 18$.

(a) Write the budget set and budget line. Find both intercepts and the slope, with x_1 on the horizontal axis.

(b) Check whether $(3, 4)$, $(6, 3)$ and $(0, 5)$ are affordable, and identify which affordable bundles exhaust the budget.

(c) In separate diagrams, show what happens when income rises to 36 at unchanged prices, and when p_1 rises to 4 at unchanged income 18 and $p_2 = 3$.

(d) Compare the original budget with prices $(4, 6)$ and income 36. Then explain why LNS implies budget exhaustion at an existing optimum.

B4. Cobb–Douglas, one step at a time · 20 points

Let $u(x) = x_1^{1/3} x_2^{2/3}$ on $\mathbb{R}^2_+$, with $p_1, p_2, m > 0$.

(a) Establish existence of an optimum using the original utility. Show that every optimum has $x_1, x_2 > 0$ before taking logarithms.

(b) On the positive domain, replace the objective by $g = \log u$. Write its Lagrangian and the first-order conditions for an interior solution with an exhausted budget.

(c) Derive demand for both goods. Verify the budget and both expenditure shares. Explain why this is the unique optimum.

(d) Calculate the bundle at $p = (2, 1)$, $m = 12$. Multiply both prices and income by 3 and check the result.

(e) Derive indirect utility $V_g(p, m)$ for the log objective and use Roy's identity to recover both demands.

(f) Find the budget multiplier for maximising g, then for maximising the original u. Explain why the bundles agree although the multipliers differ.

B5. Practising corners with Kuhn–Tucker · 20 points

Let $u(x) = 2\log(1 + x_1) + \log(1 + x_2)$, $p = (1, 1)$, $m > 0$.

(a) Write the full KKT system with nonnegative multipliers for $m - x_1 - x_2$, x_1 and x_2.

(b) First assume both goods are positive. Solve the stationarity equations and budget equation. For which incomes is this candidate strictly positive?

(c) Next set $x_2 = 0$, exhaust the budget and calculate all multipliers. For which incomes does dual feasibility hold?

(d) Combine the cases into a demand formula valid for every $m > 0$, including the transition income. Verify global optimality and uniqueness.

(e) Fill a table of $(x_1^*, x_2^*, \lambda, \mu_1, \mu_2)$ for $m = 1/2$, $m = 1$ and $m = 4$. Does $x_2^* = 0$ always imply $\mu_2 > 0$?

B6. A kink without derivatives · 10 points

Let $u(x) = \min\{x_1, 2x_2\}$, with $p_1, p_2, m > 0$.

(a) For a target utility $k \geq 0$, write the two inequalities required to attain at least k. Find the cheapest bundle attaining it and its cost.

(b) Use this cost to find the highest affordable k and derive both Marshallian demands. Explain why the chosen bundle is unique.

(c) Find demand and indirect utility at $p = (1, 2)$, $m = 12$.

(d) Why is an ordinary differentiable tangency condition inappropriate at the optimum? Explain how the fixed-proportion argument replaces it.

B7. Quasilinearity and the income boundary · 15 points

Let $u(a, q) = a + 5\log(1 + q)$, with $a, q \geq 0$, both prices equal to 1, and budget $a + q \leq m$, $m > 0$.

(a) Explain why utility is quasilinear in a and why the budget is exhausted. Substitute $a = m - q$ and state the feasible interval for q.

(b) Differentiate the reduced objective and find its unconstrained maximiser. Impose $0 \leq q \leq m$ to obtain demand at all incomes.

(c) Calculate the bundle at $m = 2$ and $m = 6$. Draw a diagram of $q^*(m)$ and explain the change of regime.

(d) Derive the indirect utility as a piecewise function of m. Check that its value and its first derivative match at the transition.

(e) Find the budget multiplier in each regime. Explain when extra income buys more q and when it buys only more a.